Closed Near-Circular Orbits Force a Power Law: A Formal Proof of the First Half of Bertrand's Theorem in Lean 4

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Abstract

Bertrand's theorem (1873) states that the only central forces for which every bounded orbit is closed are the inverse-square law and Hooke's law. All standard proofs share a first step. If every orbit near each stable circular orbit is closed, then the near-circular apsidal angle $\pi/\sqrt{3 + rV''/V'}$ must be a constant rational multiple of π, so the force is a power law $V'(r) = C\, r^{\beta^2-3}$ with $\beta \in \mathbb{Q}_{>0}$. We give a machine-checked proof of this step in Lean 4 with Mathlib, with the apsidal angle defined as an honest singular integral over the classically allowed region. The analytic core is a one-dimensional Morse lemma, proved via L'Hôpital's rule. A trigonometric substitution in Morse coordinates then turns the apsidal integral into a regular integral. Together they show that the apsidal angle is continuous in the energy up to the circular orbit, where it equals $\pi/\sqrt{U''}$. To our knowledge this is the first formalization of any part of Bertrand's theorem.

1 Introduction

Among all central forces, why do only gravity and the ideal spring produce closed orbits? Bertrand's answer [2] is a classic of mechanics, reproduced in Goldstein [5] and Arnold [1]. Rigorous treatments are surprisingly delicate. Arnold's sketch hides nontrivial analysis in its limits. Féjoz and Kaczmarek [4] give a careful local proof via Birkhoff invariants, and many elementary proofs have since appeared (e.g. [3]).

Most proofs proceed in two stages:

(I) *Near-circular orbits.* If all orbits close to stable circular orbits are closed, the force is a power law $F \propto r^{\beta^2-3}$ with rational β.

(II) *Power laws.* Among power laws, only $\beta = 1$ (Kepler) and $\beta = 2$ (harmonic oscillator) have all bounded orbits closed.

This paper formalizes stage (I) completely in Lean 4 [6]. Stage (II) and the link between closed trajectories and apsidal angles are left for future work (§6). The development is part of the repository `https://github.com/Crotalus/ai-formalized` (folder `bertrand`). It builds with Lean `v4.35.0` and Mathlib commit `56a0058`.

*AI system (model Claude Opus 5.5), developed by Anthropic. Claude designed the formalization, wrote the Lean proofs (in part via parallel sub-agents) and drafted this manuscript in collaboration with the first author.

2 Setting

A particle of unit mass moves in a central potential $V(r)$ with angular momentum $M > 0$. In the variable $x = M/r$, Binet's equation states that the radial motion, parametrized by the polar angle θ, is a one-dimensional oscillation $\frac{1}{2}x'(\theta)^2 + U(x) = E$ in the effective potential

$$U(x) = \frac{x^2}{2} + V\left(\frac{M}{x}\right).$$

The *apsidal angle*, the angle from pericentre to apocentre, is the half-period of this oscillation:

$$\Phi(E) = \int_{\{U<E\}_{x_0}} \frac{dx}{\sqrt{2(E - U(x))}},$$

where $\{U < E\}_{x_0}$ is the connected component, containing the bottom x_0 of the well, of the classically allowed region. An orbit closes after finitely many revolutions exactly when $\Phi/\pi \in \mathbb{Q}$. A circular orbit of radius r has $M^2 = r^3V'(r)$ and sits at the critical point $x_0 = M/r$ of U, with

$$U''(x_0) = 3 + \frac{rV''(r)}{V'(r)}, \tag{1}$$

and it is stable when this is positive. In Lean:

```
def apsidalAngle (U : ℝ → ℝ) (Ω : Set ℝ) (x₀ E : ℝ) : ℝ :=
  ∫ x in connectedComponentIn (Ω ∩ {x | U x < E}) x₀, 1 / √(2 * (E - U x))

def effPot (V : ℝ → ℝ) (M x : ℝ) : ℝ := x ^ 2 / 2 + V (M / x)
def angMom (V : ℝ → ℝ) (r : ℝ) : ℝ := √(r ^ 3 * deriv V r)
```

The integrand is unbounded at the turning points, so Φ is a genuinely singular (Lebesgue) integral. We do not regularize it in the definition.

3 Main theorem

Theorem 1 (Bertrand, part I)**.** *Let $0 < a < b$ and let V be C^3 on (a, b) with $V' > 0$ and $3V' + rV'' > 0$ there (every circular orbit of radius in (a, b) exists and is stable). Suppose that for every $r \in (a, b)$, every orbit with the angular momentum of the circular orbit of radius r and energy slightly above it is closed, that is, $\Phi(E) \in \pi\mathbb{Q}$. Then there is a rational $\beta > 0$ with $3 + rV''/V' = \beta^2$ on (a, b), and $V'(r) = C\, r^{\beta^2-3}$ for some $C > 0$.*

```
theorem forceLaw_of_closed_nearCircular {V : ℝ → ℝ} {a b : ℝ} (ha : 0 < a) (hab : a < b)
    (hV : ContDiffOn ℝ 3 V (Ioo a b))
    (hpos : ∀ r ∈ Ioo a b, 0 < deriv V r)
    (hstab : ∀ r ∈ Ioo a b, 0 < 3 * deriv V r + r * iteratedDeriv 2 V r)
    (hclosed : ∀ r ∈ Ioo a b, ∀ᶠ E in 𝓝[>] (circularEnergy V r),
      ∃ q : ℚ, orbitApsidalAngle V a b r E = q * π) :
    ∃ β : ℚ, 0 < β ∧ (∃ C > 0, ∀ r ∈ Ioo a b, deriv V r = C * r ^ ((β : ℝ) ^ 2 - 3)) ∧
      ∀ r ∈ Ioo a b, 3 + r * iteratedDeriv 2 V r / deriv V r = (β : ℝ) ^ 2
```

The orbit's allowed region is taken inside the radii (a, b), i.e. $\Omega = (M/b, M/a)$ in the x variable.

Proof outline. Fix r and the well U of its circular orbit.

1. *Morse chart* (§4). There is a C^1 map X with $X(0) = x_0$, $X' > 0$ and $U(X(s)) = U(x_0) + s^2/2$ near 0, with $X'(0) = 1/\sqrt{U''(x_0)}$.

2. *Regularization* (§5). For $0 < \varepsilon$ small, $\Phi(U(x_0) + \varepsilon) = \int_{-\pi/2}^{\pi/2} X'(\sqrt{2\varepsilon}\sin\theta)\, d\theta =: F(\varepsilon)$. The function F is continuous on $[0, \rho^2/2)$, with $F(0) = \pi X'(0) = \pi/\sqrt{U''(x_0)}$.

3. *Rationality.* By hypothesis $F(\varepsilon)/\pi \in \mathbb{Q}$ for small $\varepsilon > 0$. A continuous rational-valued function on an interval is constant, since otherwise its image would contain an irrational number. So F is constant near 0^+, and by continuity $F(0) \in \pi\mathbb{Q}$. Hence $\sqrt{U''(x_0)} \in \mathbb{Q}$.

4. *Power law.* By (1), $r \mapsto \sqrt{3 + rV''/V'}$ is continuous on (a, b) and rational-valued, hence a constant β. Then $rV'' = (\beta^2 - 3)V'$, so $(V'(r)\, r^{3-\beta^2})' = 0$ and $V' = C\, r^{\beta^2 - 3}$.

Step 3 is where continuity of Φ *in the energy* is indispensable. A function with values in $\pi\mathbb{Q}$ can perfectly well have an irrational limit. This is why the near-circular formula $\Phi \to \pi/\sqrt{U''}$, which textbooks derive by linearization, does not by itself prove the step.

4 The one-dimensional Morse lemma

```
theorem exists_morseChart {U : ℝ → ℝ} {Ω : Set ℝ} {x₀ : ℝ} (hΩ : IsOpen Ω) (hx₀ : x₀ ∈ Ω)
    (hU : ContDiffOn ℝ 3 U Ω) (h1 : deriv U x₀ = 0) (h2 : 0 < iteratedDeriv 2 U x₀) :
    ∃ c : MorseChart U Ω x₀, deriv c.X 0 = 1 / √(iteratedDeriv 2 U x₀)
```

The usual proof writes $U(x) - U(x_0) = (x - x_0)^2 g(x)$ with $g(x) = \int_0^1 (1-t)U''(x_0 + t(x - x_0))\, dt$, and needs differentiation under the integral sign. The formal proof avoids this. It sets $q(x) = (U(x) - U(x_0))/(x - x_0)^2$ and $s(x) = (x - x_0)\sqrt{2q(x)}$, and uses Mathlib's L'Hôpital rule (`lhopital_zero_nhdsNE`) to get $q(x) \to U''(x_0)/2$. Hence $s'(x_0) = \sqrt{U''(x_0)}$. Away from x_0, differentiating $s^2 = 2(U - U(x_0))$ gives $s' = U'/s$, which tends to $\sqrt{U''(x_0)}$ as well. So s is C^1 with positive derivative near x_0, and the local inverse given by Mathlib's inverse function theorem (`HasStrictFDerivAt.toOpenPartialHomeomorph`) is the chart X.

5 The apsidal angle in a Morse chart

```
theorem MorseChart.apsidalAngle_eq (c : MorseChart U Ω x₀) {ε : ℝ} (hε : 0 < ε)
    (hερ : ε < c.ρ ^ 2 / 2) :
    apsidalAngle U Ω x₀ (U x₀ + ε) =
      ∫ θ in -(π / 2)..π / 2, deriv c.X (√(2 * ε) * sin θ)
```

Let $\sigma = \sqrt{2\varepsilon}$. First, the allowed component equals $X((-\sigma, \sigma))$. This image is connected, lies in $\{U < E\}$ and contains x_0. Conversely, the component is an interval around x_0 that cannot contain the turning points $X(\pm\sigma)$, where $U = E$. Second, the map $\theta \mapsto X(\sigma\sin\theta)$ is a C^1 bijection from $(-\pi/2, \pi/2)$ onto that component. On it $2(E - U) = \sigma^2\cos^2\theta$, so the Jacobian $X'(\sigma\sin\theta)\sigma\cos\theta$ cancels the singularity exactly. The change of variables uses Mathlib's Lebesgue version `integral_image_eq_integral_abs_deriv_smul`, which requires neither continuity nor integrability of the singular integrand. Continuity of $F(\varepsilon) = \int X'(\sqrt{2\varepsilon}\sin\theta)\, d\theta$ on $[0, \rho^2/2)$ follows from continuity of parametric integrals with jointly continuous integrand.

6 Discussion

Size and checking. The development consists of five files, about 750 lines in all. It has no `sorry`, and `#print axioms` on the main theorem reports only `propext`, `Classical.choice` and `Quot.sound`. The proof was developed in parallel. We first fixed a small shared interface: the definition of `apsidalAngle` and a `MorseChart` structure. Separate agents then proved the Morse lemma and the chart computation against it, while the physics layer and the assembly were written in parallel.

Stage II. With $V' = Cr^{\beta^2-3}$ on all of $(0,\infty)$ and all bounded orbits closed, the same continuity-plus-rationality argument, applied over the connected range of bound energies, shows that $\Phi(E) = \pi/\beta$ for every bound energy. Comparing with the limit $E \to \infty$ for $\beta^2 > 2$ (where $\Phi \to \pi/2$) and $E \to 0^-$ for $\beta^2 < 2$ (where $\Phi \to \pi/\beta^2$) leaves $\beta = 2$ and $\beta = 1$. Formalizing this needs a global Morse chart for a single-well potential and two asymptotic evaluations of the apsidal integral. The regularized integral of §5 is designed to make both tractable.

Trajectories. We take as given that an orbit closes exactly when $\Phi \in \pi\mathbb{Q}$. Deriving this from Newton's equations (Binet's equation and periodicity of the radial motion) would connect the result to Physlib's central-force library.

Author contributions. N.L. chose the problem, directed the work and reviewed the results. Claude, an AI system developed by Anthropic, designed the formal statements and proof architecture, wrote the Lean proofs (delegating the Morse lemma and the chart computation to sub-agents) and drafted the manuscript. All statements were checked by the Lean kernel. As an AI system cannot take responsibility for a publication, N.L. is accountable for its content.

References

[1] V. I. Arnold. *Mathematical Methods of Classical Mechanics*, volume 60 of *Graduate Texts in Mathematics*. Springer, 2nd edition, 1989.

[2] Joseph Bertrand. Théorème relatif au mouvement d'un point attiré vers un centre fixe. *Comptes Rendus de l'Académie des Sciences de Paris*, 77:849–853, 1873.

[3] Siu A. Chin. A truly elementary proof of Bertrand's theorem, 2014.

[4] Jacques Féjoz and Laurent Kaczmarek. Sur le théorème de Bertrand (d'après Michael Herman). *Ergodic Theory and Dynamical Systems*, 24, 2004.

[5] Herbert Goldstein, Charles Poole, and John Safko. *Classical Mechanics*. Addison-Wesley, 3rd edition, 2002.

[6] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.